1.

a) (3 pts)Considering that the lossless TL has no electrical connection to any elements at either end, each resonant mode (or standing wave) will have a resonance frequency

$$f = \frac{v}{2\ell} n,$$

where $n \geq 1$. Therefore, the resonant frequencies will be

$$f = \frac{10^8}{2 \times 10} n = 5n \,\text{MHz}.$$

b) (4 pts)The general voltage expression for a TL is given by

$$\tilde{V}(z) = V^+ e^{-j\beta z} + V^- e^{j\beta z} \tag{1}$$

whereas the general current expression is

$$\tilde{I}(z) = \frac{V^+}{Z_o} e^{-j\beta z} - \frac{V^-}{Z_o} e^{j\beta z}.$$

Since both ends of the transmission line are open circuited, the current at these locations must be zero. Then, at $z = 0$ we find that

$$V^+ = V^-. \tag{2}$$

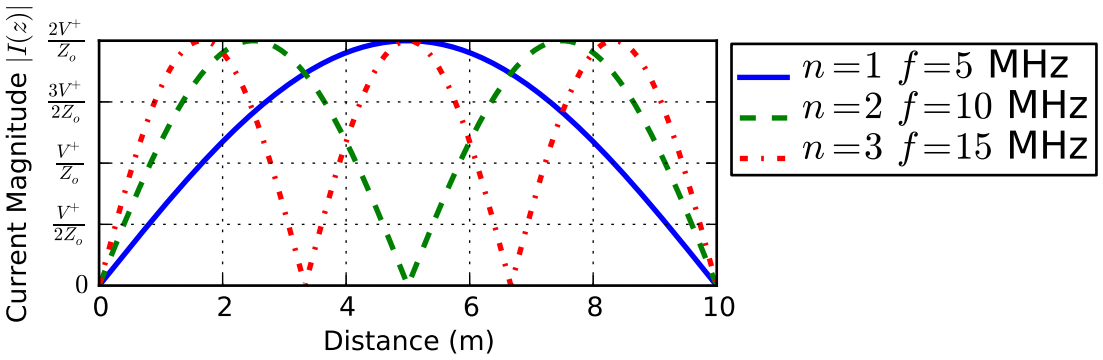
Therefore, the general current phasor takes the form of

$$\tilde{I}(z) = \frac{V^+}{Z_o} e^{-j\beta\ell} - \frac{V^+}{Z_o} e^{j\beta\ell} = -\frac{V^+}{Z_o} 2j \sin(\beta z)$$

for a lossless TL which is open circuited at both ends. Knowing that $\beta = \frac{2\pi}{\lambda}$ and $\lambda = \frac{v}{f} = \frac{2\ell}{n}$, we can write

$$\tilde{I}(z) = -\frac{V^+}{Z_o} 2j \sin(\frac{n\pi}{\ell} z) \,\text{A}.$$

The current magnitude is sketched in the below figure.

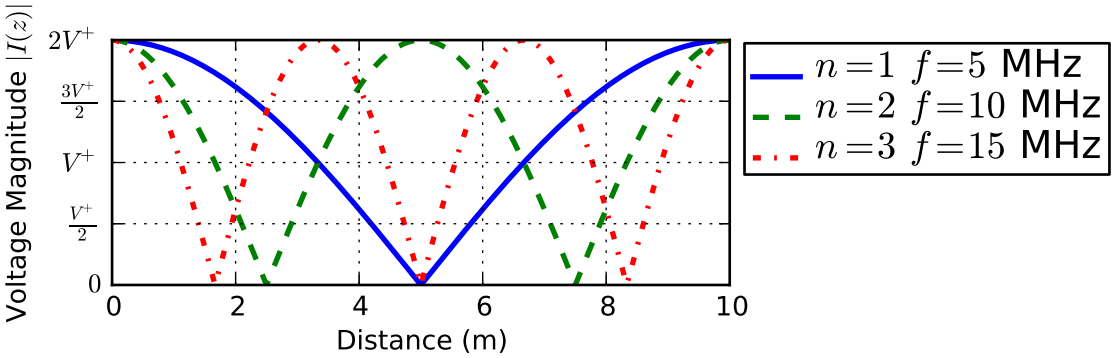


As you may notice, each mode n has n half wavelenghts fitted into the length of the TL, ℓ and the higher modes oscillate with higher frequencies.

c) (3 pts)Using (1) and (2), we can also write

$$\tilde{V}(z) = 2V^+ \cos(\frac{n\pi}{\ell} z)\,\mathrm{V},$$

from which we obtain the following figure.



The comments mentioned related to the figure in part (b) are also valid for this figure. Additionally, comparing this figure with that in part (b), one can say that the voltage and current nulls of the resonant standing wave modes on this TL are seperated by a quarter wavelength.

2.

a) (1 pt)The signal wavelength λ is $\lambda = \frac{v}{f} = \frac{2\times10^8}{\pi\times10^8/2\pi} = 4\,\mathrm{m}$.

b) (1 pt)Since the end of the line (i.e. at $d = 0$) is open circuited, $I(0) = 0$.

c) (1 pt)Yes, because the magnitude of the reflection coefficient at the open termination is one (i.e., $|\Gamma_L| = 1$), then, the reflected wave has the same amplitude as the incident wave, which is enough to set up a standing wave.

d) (1 pt)If $I_R = 2\,\mathrm{A}$ then the current in the transmission line at $d = l$ is zero (i.e., $I(l) = 0$). Since successive current nulls are $\lambda/2$ apart, the smallest non-zero transmission line length is $l = \lambda/2 = 2\,\mathrm{m}$.

e) (1 pt)If $l = \frac{\lambda}{2}$ then the voltage at $d = l/2$ is zero (i.e, $V(\frac{l}{2}) = 0$), because half-way between successive current nulls there is a voltage null.

f) (1 pt)If $l = \frac{\lambda}{2}$ then the current at $d = l/2$ can not be zero (i.e., $I(\frac{l}{2}) \neq 0$), because at this point the current is a maximum.

g) (1 pt)If $I_R = 0$ then the voltage at $d = l$ is also zero (i.e., $V(l) = 0$). Since voltage and current nulls are $\lambda/4$ apart, the smallest non-zero transmission line length is $l = \frac{\lambda}{4} = 1\,\mathrm{m}$.

h) (1 pt)If $I(\frac{l}{2}) = 0$ and $I(0) = 0$ then $\frac{l}{2} = \frac{\lambda}{2}$ because the separation between two successive current nulls is $\lambda/2$ (non-trivial case). Therefore, $l = \lambda = 4\,\mathrm{m}$.

i) (2 pt)In part (h), $I(\frac{l}{2}) = 0$ and $l = \lambda$ which implies that $I(l) = 0$. Since $I_R = 2\angle 0°\,\mathrm{A}$ then $V(l) = I_R \times 50\,\Omega = 100\angle 0°\,\mathrm{V}$.

3.

a) (5 pts)Since the voltage measured at any distance along the TL is given by

$$V(d) = V^+(e^{j\beta d} + \Gamma_L e^{-j\beta d}),$$

where $\Gamma_L = \frac{Z_L - Z_o}{Z_L + Z_o}$, the voltages at one end is

$$V_L = V(0) = V^+(1 + \Gamma_L)$$

whereas the input voltage of the quarter-wave section at the other end is

$$V_{in} = V(\frac{\lambda}{4}) = jV^+(1 - \Gamma_L).$$

Therefore, we can write

$$\frac{V_L}{V_{in}} = -j\frac{1 + \Gamma_L}{1 - \Gamma_L} = -j\frac{Z_L}{Z_o},$$

from which we can obtain

$$I_L = \frac{V_L}{Z_L} = -j\frac{V_{in}}{Z_o} = -j\frac{100}{50} = -2j\,\text{A}.$$

b) (2 pts)If the load end is open then $I_L = 0$. This implies that the voltage input must also be zero ($V_{in} = jZ_oI_L = 0$), and therefore, $V_{in} = 100\angle 0^\circ\,\text{V}$ is not a possibility in this situation.

c) (3 pts)Referring to page 3 of Lecture 33, the input current is given by

$$I_{in} \equiv I(\frac{\lambda}{4}) = \frac{jV^+ + jV^-}{Z_o} = j\frac{V(0)}{Z_o} = j\frac{V_L}{Z_o}$$

From part (b), we know that $V_L = -j\frac{Z_L}{Z_o}V_{in}$ which yields

$$I_{in} = \frac{V_{in}}{Z_{in}} = \frac{Z_L}{Z_o^2}V_{in} \quad \Rightarrow \quad I_{in} = 4\,\text{A}.$$

4.

a) (3 pts)We are given the voltage at the load (i.e. $V_L = j15 = 15e^{j\frac{\pi}{2}}$ where $|V_L| = 15\,\text{V}$) and the load resistance so the power dissipated at the load is simply calculated as

$$P_L = \frac{1}{2}\text{Re}\{V_L I_L^*\} = \frac{1}{2}Re\{V_L \cdot \frac{V_L^*}{Z_L^*}\} = \frac{1}{2}\cdot\frac{|V_L|^2}{R_L} = 0.75\,\text{W}.$$

b) (7 pts)Even though the length of the line is $\ell = \frac{3}{4}\lambda$, it can be divided into $\frac{\lambda}{2}$ and $\frac{\lambda}{4}$ sections in order to benefit from the advantages of half-wave and quarter-wave transformers. Thus, the voltage at a distance $\frac{\lambda}{2}$ from the load is simply given by

$$V(-\frac{\lambda}{2}) = -V_L = -j15\,\text{V},$$

whereas the current is

$$I(-\frac{\lambda}{2}) = -I_L = -j0,1\,\text{A}.$$

Now, applyling $\frac{\lambda}{4}$ transformer concepts for the rest of the line, we will get

$$V_{in} = V(-\frac{3}{4}\lambda) = jI(-\frac{\lambda}{2})Z_o = 7.5\,\mathrm{V},$$

and current at the input of the line will be

$$I_{in} = I(-\frac{3}{4}\lambda) = j\frac{V(-\frac{\lambda}{2})}{Z_o} = 0.2\,\mathrm{A}.$$

The current I_{in} is the same as the generator current I_g. The generator voltage can be found by

$$\frac{V_g - V_{in}}{Z_g} = I_g \quad \Rightarrow V_g = 15\,\mathrm{V}$$

Then, we can find the power delivered by the generator

$$P_g = \frac{1}{2}Re\left\{V_g I_g^*\right\} = 1.5\,\mathrm{W}.$$